

Satellite Communications

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“Satellite Communications” - Performance Evaluation of Computer Networks And Systems

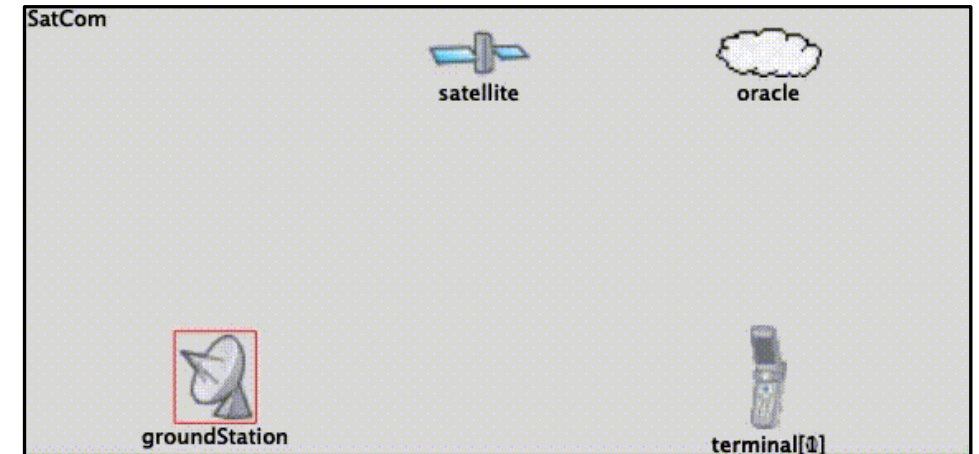


Introduction

- **System overview:**

- N terminals, each with a dedicated FIFO queue at the Ground Station
- Packets:
 - Uniformly distributed sizes
 - Exponentially distributed inter-arrival times
- Communication slots of fixed size (80 ms)
- **MaxCR scheduling algorithm**

Coding Rate	Capacity (bytes)
L3	904/M
L2	1356/M
L1	1808/M
R	2260/M
H1	2712/M
H2	3164/M
H3	3616/M

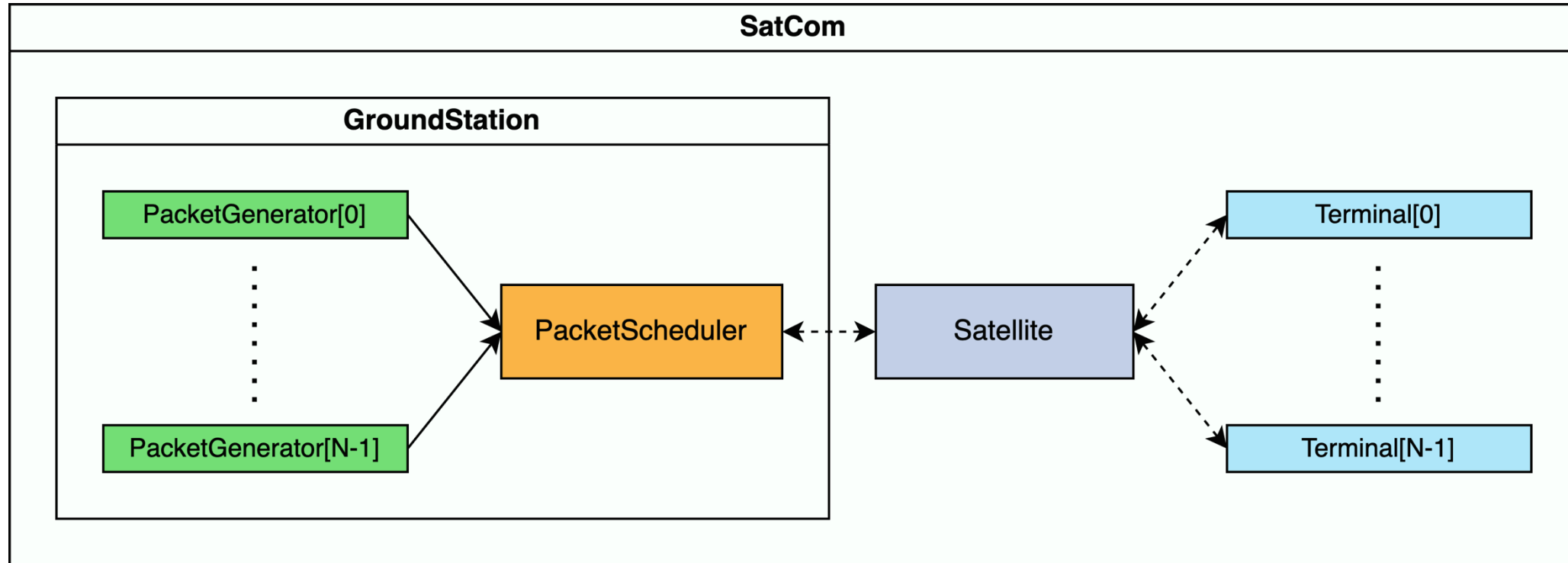


KPIs, Parameters and Factors

- Key Performance Indicators:
 - **Throughput**
 - **Mean Delay**
 - **Mean Frame Utilization**
- Parameters:
 - **minPacketSize** and **maxPacketSize**
 - **meanPacketInterarrivalTime**
- Factors:
 - **N** , the number of terminals
 - **M** , the number of blocks per frame
 - The distribution of the **coding rates**

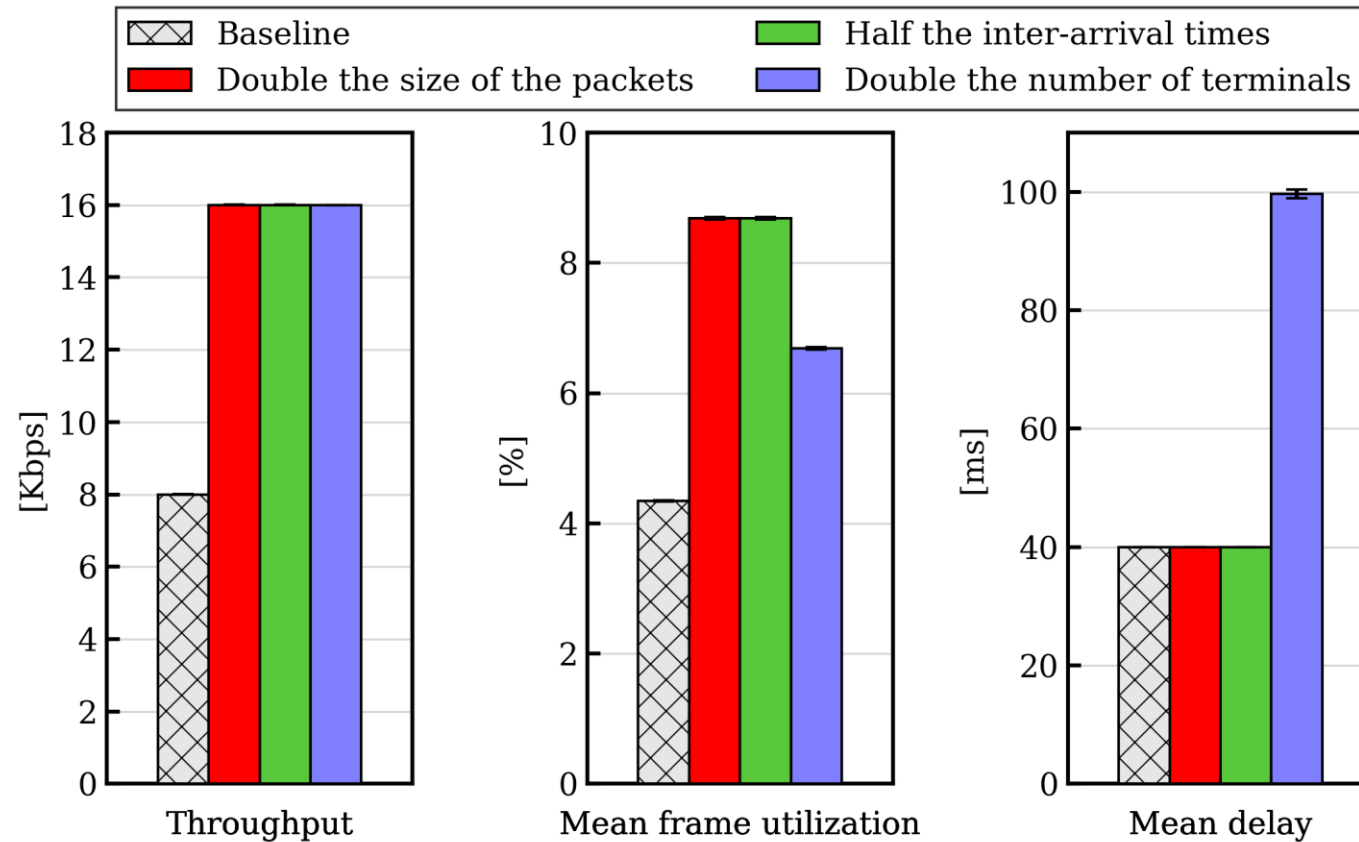


Implementation



OMNeT++ network design. Dashed arrows indicate direct (wireless) connections.

Verification: Consistency test



Consistency test



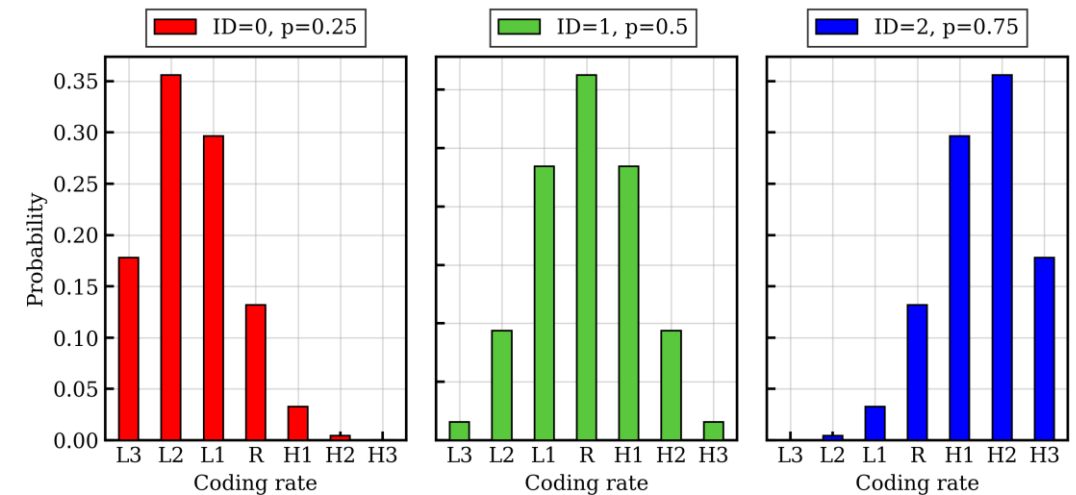
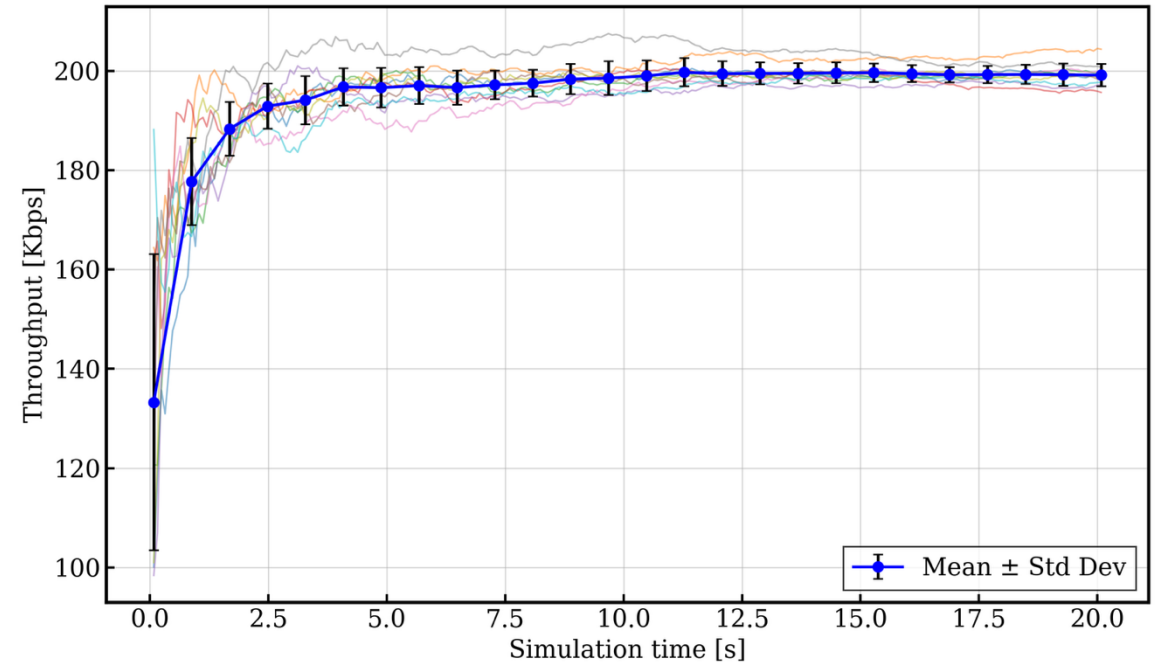
Calibration

- Calibrated parameters:

Parameter	Value
minPacketSize	20 bytes
maxPacketSize	80 bytes
meanPacketInterarrivalTime	40 ms
warm-up-period	10 s
sim-time-limit	800 s

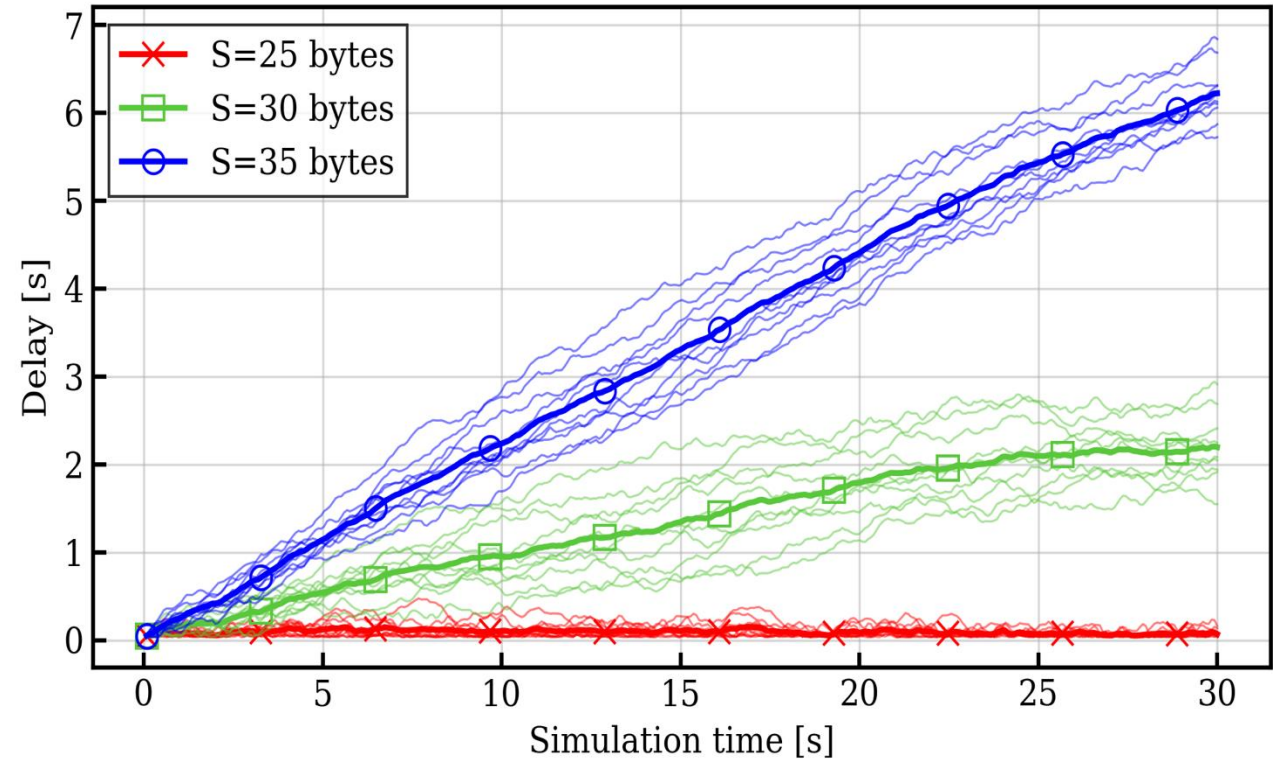
- Coding rate distributions:

- (discretized) **Uniform** (0, 7)
- (discretized) **Normal** ($\mu = 3.5$, $\sigma = 1$)
- Binomial** ($n = 6$, $p = \frac{\text{terminalId} + 1}{N + 1}$)

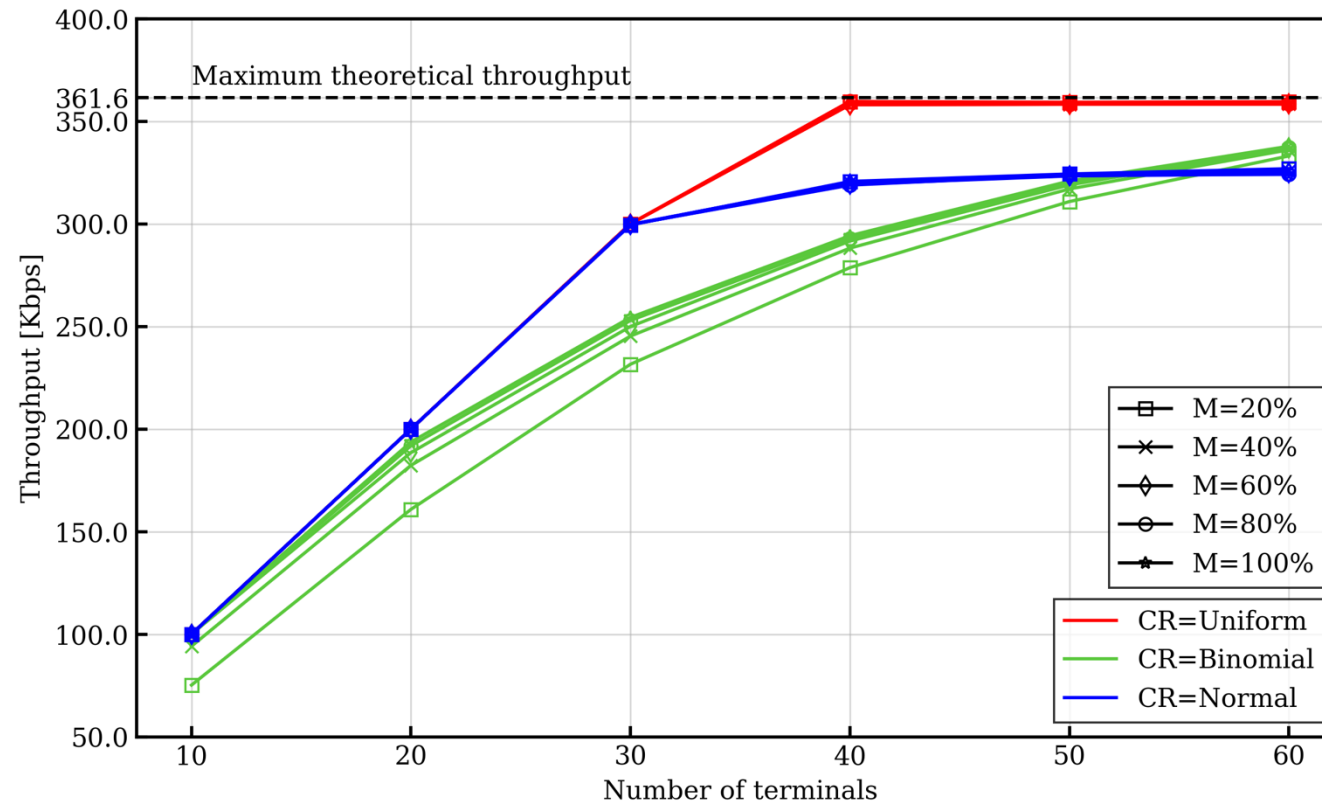


Experiment design

Factor	Value
N	10, 20, 30, 40, 50, 60
M	20%, 40%, 60%, 80%, 100%
CR	Uniform, Normal, Binomial



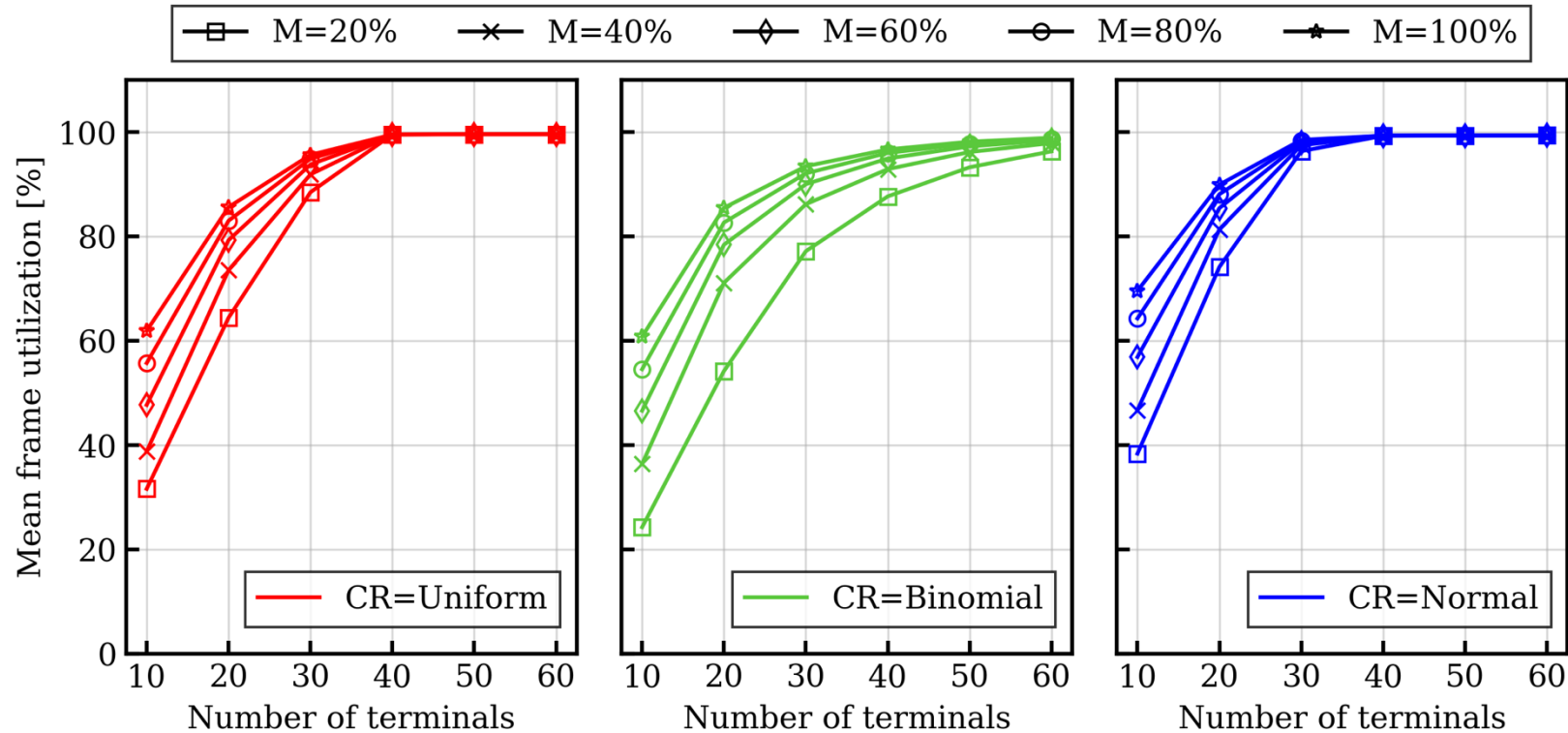
Data analysis: Throughput



Throughput of the system as a function of number of terminals, the number of blocks per frame and the distribution of the coding rates

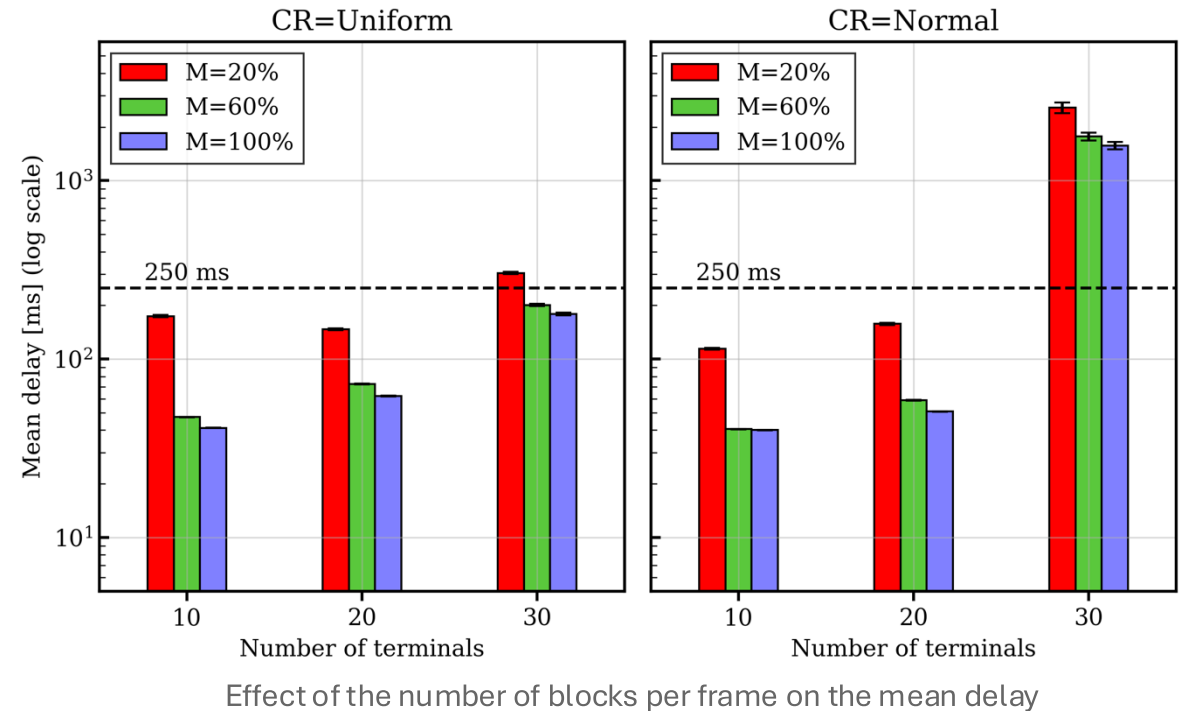
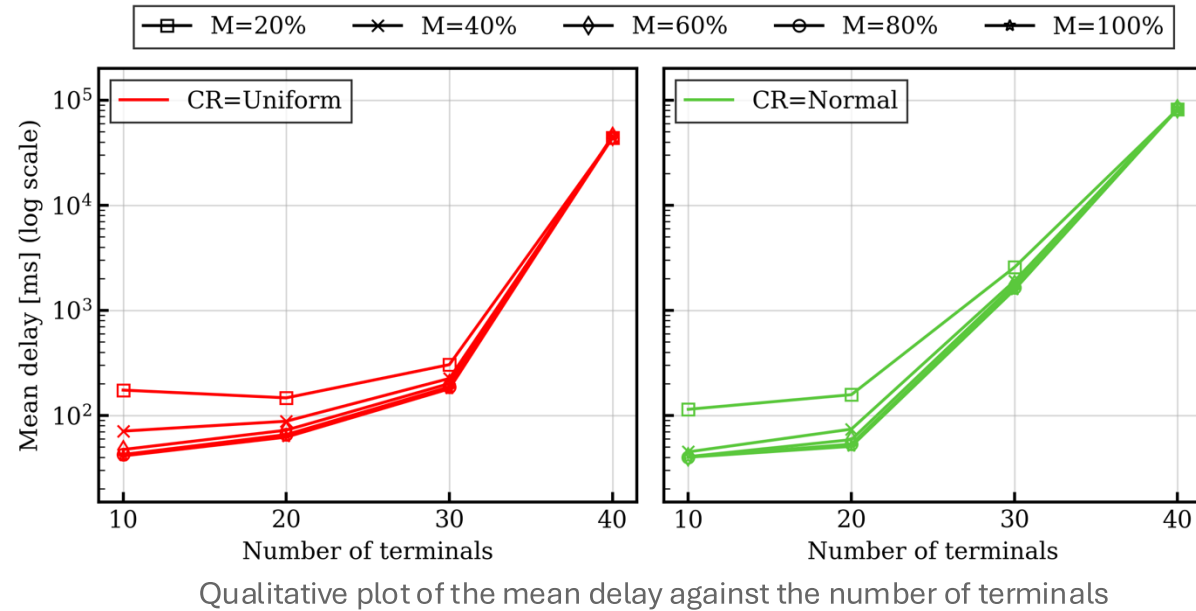


Data analysis: Mean Frame utilization

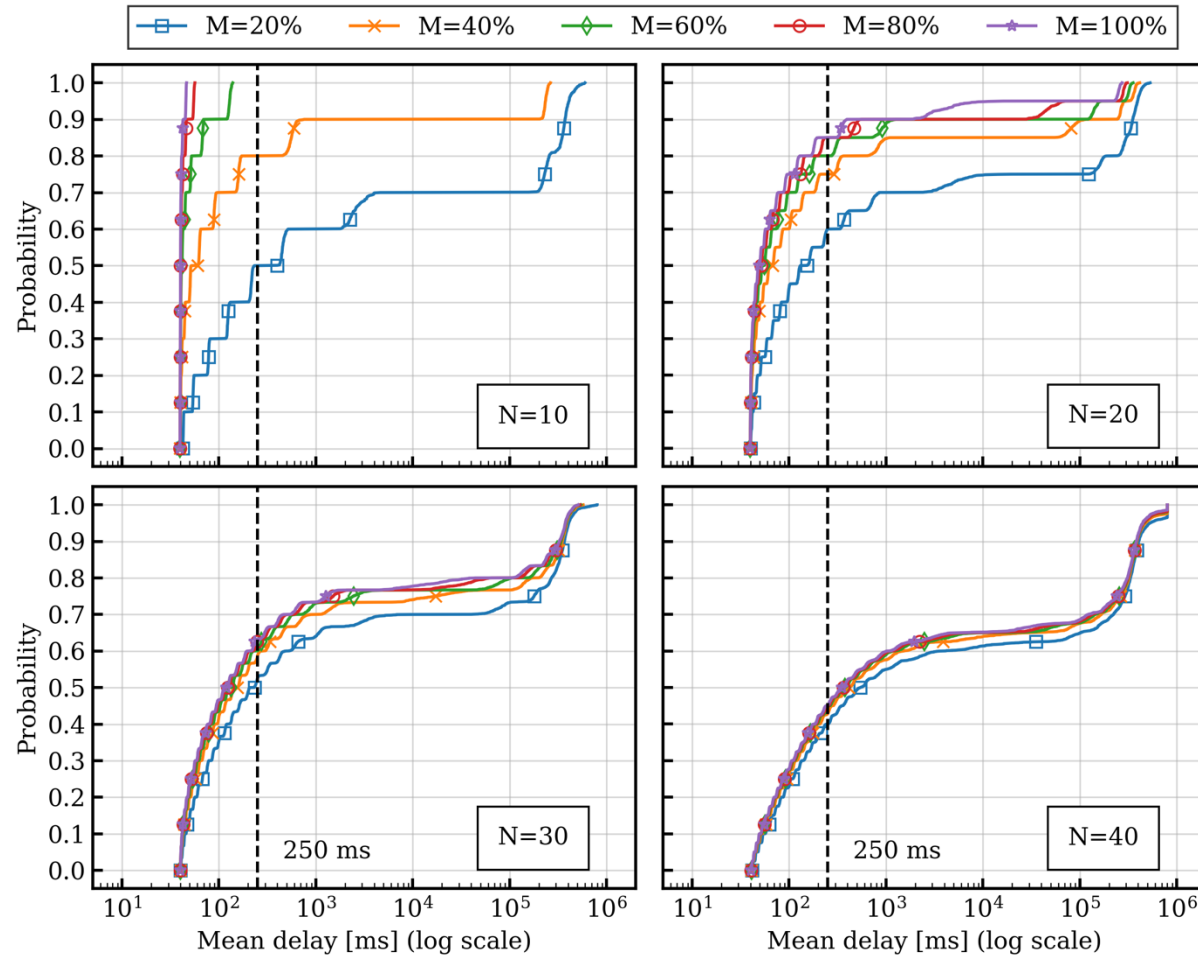


Mean Frame Utilization as a function of number of terminals, the number of blocks per frame and the distribution of the coding rates

Data analysis: Mean Delay (Uniform and Normal)



Data analysis: Mean Delay (Binomial)



ECDF of the mean delays of the terminals when the coding rates are binomially distributed